For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 3

6031/3

NOVEMBER 2021 SESSION 2 hours 30 minutes

Additional materials:
Answer paper
Data Booklet
Electronic calculator
Graph paper

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 14 printed pages and 2 blank pages.

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Section A

*Answer any **two** questions from this section.*

1 **(a)** **(i)** Write the electronic configuration of Ca^{2+} ion.

(ii) Sketch a graph to show the variation in the metallic radii of Group (II) metals, Magnesium to Barium.

(iii) Explain the trend in the ease of formation of Group (II) cations.

[6]

(b) **Table 1.1** shows values of the lattice energies and solubilities of sulphates of Group (II) elements.

Table 1.1

compound	lattice enthalpy/ $kJmol^{-1}$	solubility/moles per 100 g of water
$MgSO_4$	−91.20	$3\,600 \times 10^{-4}$
$CaSO_4$	+17.80	11×10^{-4}
$SrSO_4$	+18.70	062×10^{-4}
$BaSO_4$	+19.40	0.09×10^{-4}

(i) Draw an enthalpy diagram to show the energy changes that take place when magnesium sulphate dissolves in water.

(ii) Explain the reasons for the variation in the solubilities of the Group (II) sulphates.

(iii) Suggest a laboratory application for the low solubility of $BaSO_4$.

[5]

(c) Explain why

(i) calcium sulphate is a compound of medical importance,

(ii) calcium hydroxide is used in agriculture.

[4]
[Total: 15]

2 (a) **Fig. 2.1** shows the melting point of Period 3 oxides, **A** to **G**.

[Letters **A** to **G** represent the oxides of the successive elements and are not chemical formulae.]

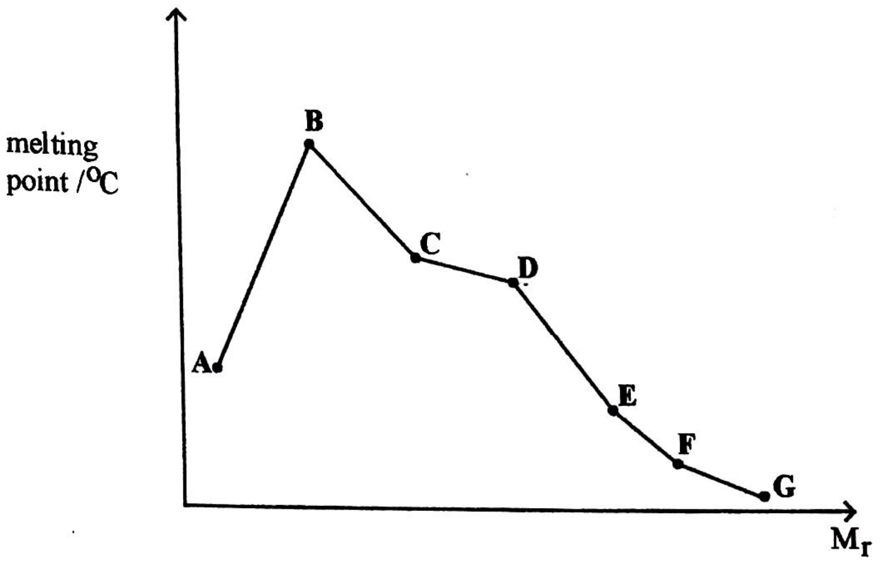


Fig. 2.1

(i) Name the lattice structure of **A**.

(ii) Explain why

1. the melting point rises sharply from **A** to **B**,

2. **D** has a higher melting point than **E**,

3. **C** only conducts electricity in molten state.

[5]

(b) Covalent bonds are formed between carbon atoms resulting in long chain and ring structures.

(i) State the importance of this carbon property,

(ii) Explain the carbon's ability to show this property.

[5]

(c) In an experiment, Pb_3O_4, was reacted with dilute nitric acid. The mixture was filtered and a dark brown solid separated.

(i) Write a balanced equation for the reaction that occured.

(ii) Explain how the reaction shows that PbO is a stronger base than PbO_2.

(iii) Galena, an impure ore of PbO contains 56% lead sulphide, PbS. 73 g of PbO was produced when 180 g of glena were heated in air.

$$2PbS + 2O_2 \rightarrow 2PbO + 2SO_2$$

Calculate

1. mass of lead sulphide in glena,

2. % yield of PbO.

(iv) Explain how the extraction of lead oxide from glena negatively affects the environment.

[5]
[Total: 15]

3 (a) State Le Chatelier's principle. [1]

(b) Gases A and B react according to the equation:

$$A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}.$$

Table 3.1 shows the partial pressures obtained for the reaction at 500 °C.

Table 3.1

	partial pressure/Pa	
	initial	equilibrium
A	7.27×10^6	3.41×10^6
B	4.22×10^6	////////////////////////////
AB	0	7.72×10^6

(i) Deduce the partial pressure of B, at equilibrium.

(ii) Calculate the K_p for the reaction.

[3]

(c) Nitrogen reacts with hydrogen according to the equation

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \quad \Delta H_r = -92 \text{ kJmol}^{-1}.$$

(i) Explain how the reaction can be made to be as economic as possible for an industrial process.

(ii) State any **one** environmental problem that is associated with the use of nitrate fertilisers and explain its impact on the environment.

[6]

(d) (i) Define *common ion effect* as applied to ionic equilibria.

(ii) Explain why sodium chloride is added to the sodium stearate, $C_{17}H_{35}COONa$, solution during the manufacture of soap.

(iii) Calculate the solubility product of sodium stearate, given that the solubility of sodium stearate is 5×10^{-4} moldm^{-3}.

[5]
[Total: 15]

[Turn over

3187

3184

Section B

*Answer any **one** question from this section.*

4 (a) Chlorine gas disproportionates to a mixture of chlorate (I) and chlorate (V) in alkaline conditions.

(i) Define the term *disproportionation.*

(ii) Write an ionic equation for the disproportionation of Cl_2 in

1. cold $NaOH_{(aq)}$

2. hot $NaOH_{(aq)}$

[5]

(b) (i) Describe and explain the trend in the reactivity of halogens with hydrogen gas.

(ii) State, with a reason, the hydrogen halide that produces the weakest acid when dissolved in water.

[5]

(c) Explain why

(i) $Cl - Cl$ bond energy is higher than the $F - F$ bond energy,

(ii) electronegativity decreases down the Group (VII) elements,

(iii) HCl can be prepared by the action of $H_2SO_{4(aq)}$ on solid $NaCl$ whereas HBr cannot be prepared by the same method.

[5]
[Total: 15]

5 (a) (i) State and explain the variation in the stability of the +4 oxidation state of Group (IV) elements.

(ii) Comment on the stability of CO_2 and CO.

[5]

(b) Magnesium and aluminium react with chlorine on heating to form their respective chlorides.

(i) Write equations for the formation of the chlorides.

(ii) Describe and explain the differences in the reactions of the chlorides with water.

(iii) Comment on the acid/base nature of the chlorides.

[5]

(c) (i) State and explain the trend in the thermal decomposition of Group (II) carbonates.

(ii) Describe and explain what happens when excess $CO_{2(g)}$ is bubbled through lime water.

[5]
[Total: 15]

Section C

Answer any two questions from this section

6 **(a)** Carbon-carbon double bonds in benzene are not typical carbon-carbon double bonds.

(i) Describe and explain any two chemical reactions that can be used to prove this observation.

(ii) Explain why benzene is generally inert.

[4]

(b) **Fig. 6.1** shows the structure of estradiol.

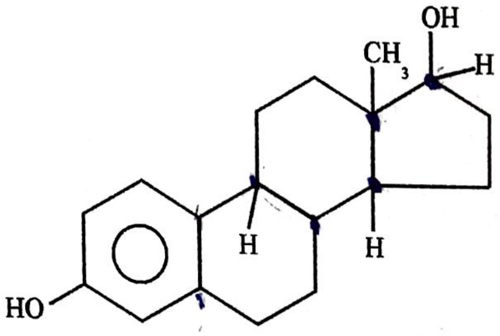


Fig. 6.1

(i) State

1. the number of chiral centres in estradiol,

2. with reasons, the acid base nature of estradiol.

(ii) Explain why estradiol

1. readily dissolves in water,

2. has a high boiling point.

(iii) Draw the structure of the organic product formed when estradiol reacts with

1. ethanoyl chloride,

2. aqueous bromine in the dark.

(iv) State the observations made in each of the reactions in **(iii)**.

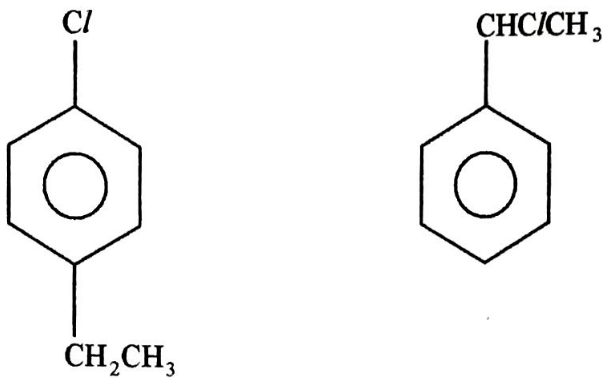
[9]

(c) Suggest methods that can be used to reduce the destruction of the ozone layer through pollution by chlorine containing compounds.

[2]

[Total: 15]

7 **(a)** **Fig. 7.1** shows the structures of two organic compounds **A** and **B**.



A B

Fig. 7.1

(i) State the type of isomerism shown by **A** and **B**.

(ii) Compare the reactions of **A** and **B** with $NaOH_{(aq)}$.

[4]

(b) The emperical formula of an organic compound **R** is CH_2O. Chemical tests carried out on **R** and the observations made are shown in **Table 7.1**.

Table 7.1

test	**observation**
R was warmed with alkaline iodine	yellow precipitate formed
2,4-DNPH was added to **R**	no observable changes
Na_2CO_3 was added to **R**.	bubbles of gas produced

(i) Deduce the functional groups in **R**.

(ii) Further chemical tests carried out on **R** showed that **R** is a chiral molecule and has a molecular mass of 90.08.

Deduce the structural formula of **R**.

[7]

(c) **Fig. 7.2** shows some of the reactions of 1,4 –dichlorobutane.

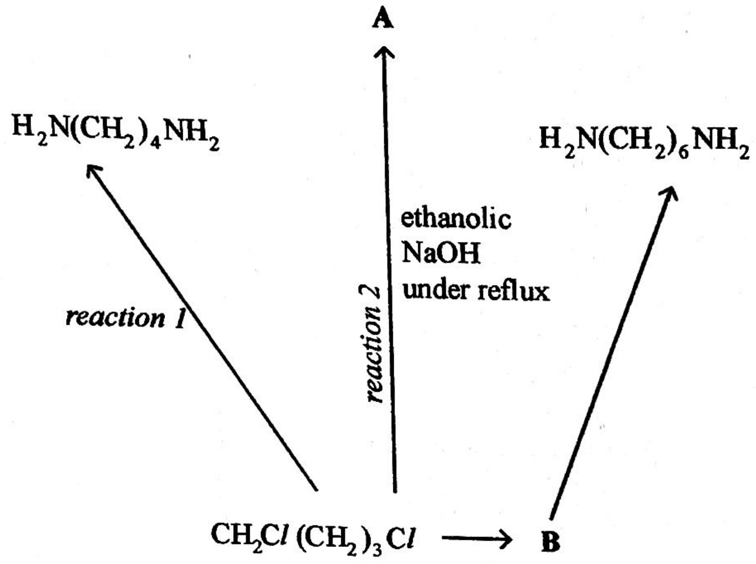


Fig. 7.2

(i) Name

1. *reaction 1,*

2. *reaction 2.*

(ii) Draw the structural formula of **A**.

(iii) Deduce the conditions needed for the formation of **B**.

[4]
[Total:15]

8 An organic compound, with structural formula $CH_3COCH_2CH_2COOCH_3$ undergoes the reaction scheme shown in **Fig. 8.1**.

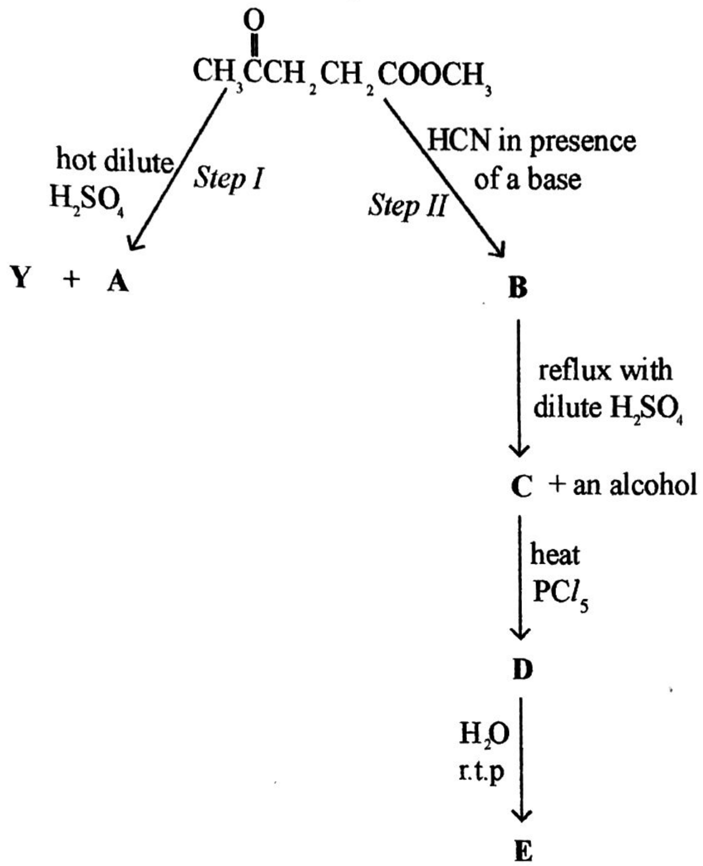


Fig. 8.1

(a) **(i)** Draw the structural formula for **A**, **Y** and **B**.

(ii) Name the type of reaction undergone in

1. step I,
2. step II.

[5]

(b) Write the reaction mechanism for *step II*. [3]

(c) Suggest the structural formula for

(i) **C**,

(ii) **D**,

(iii) **E**.

[3]

(d) **(i)** State, with reasons, the acid/base nature of **E**.

(ii) Comment on the use of solution **E** as a component of tear gas used by law enforcement agencies.

[4]
[Total:15]

Section D

*Answer any **one** question from this section*

9 **(a)** **(i)** State the

1. formula of the iron containing compound present in haematite,

2. oxidation state of iron in haematite.

(ii) Give an equation for the redox reaction used in the extraction of iron from its ores.

[3]

(b) Compare the stability of Fe^{2+} and Fe^{3+} in the presence of

(i) sodium hydroxide solution,

(ii) potassium cyanide.

[Use of data from the data booklet is recommended]

[4]

(c) When a protein sample is digested the resulting, amino acids can be separated using ion exchange chromatography.

(i) Describe ion exchange chromatography.

(ii) State the effect of pH on ion exchange chromatography for the separation of amino acids.

[5]

(d) **(i)** State any **one** disadvantage associated with the use of fossil as a source of energy.

(ii) Suggest **two** alternative energy sources.

[3]
[Total: 15]

10 **(a)** **(i)** Explain the term *nanomaterial*.

(ii) Explain why nanoparticles are more toxic than larger particles.

[3]

(b) **(i)** State the harmful effects of disposing untreated industrial waste into the sea.

(ii) Outline **three** problems associated with land filling.

(iii) Describe the treatment of industrial waste using ion exchange.

[8]

(c) State and explain the observations that take place when concentrated hydrochloric acid is added to a solution of Cu^{2+} ions.

[4]

[Total: 15]

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